

IIIT B COMET FOUNDATION BANGALORE

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Task 1.2



Seventh International Olympiad, 1965

1965/1

Determine all values x in the interval $0 \leq x \leq 2\pi$ which satisfy the inequality

$$2 \cos x \leq \left| \sqrt{1 + \sin 2x} - \sqrt{1 - \sin 2x} \right| \leq \sqrt{2}.$$

1965/2

Consider the system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0$$

with unknowns x_1, x_2, x_3 . The coefficients satisfy the conditions:

- (a) a_{11}, a_{22}, a_{33} are positive numbers.
- (b) The remaining coefficients are negative numbers.
- (c) In each equation, the sum of the coefficients is positive.

Prove that the system has only the trivial solution $x_1 = x_2 = x_3 = 0$.

1965/3

Given a tetrahedron $ABCD$ whose edges AB and CD have lengths a and b respectively. The distance between the skew lines AB and CD is d , and the angle between them is ω . The tetrahedron is divided by a plane ε , parallel to lines AB and CD . The ratio of distances of ε from AB and CD is k . Compute the ratio of volumes of the two solids obtained.

1965/4

Find all sets of four real numbers x_1, x_2, x_3, x_4 such that the sum of any one and the product of the other three is equal to 2.

1965/5

Consider $\triangle OAB$ with acute angle AOB . Through a point M , perpendiculars are drawn to OA and OB , with feet P and Q respectively. The intersection of the altitudes of $\triangle OPQ$ is H . Find the locus of H when M varies over:

1. the side AB
2. the interior of $\triangle OAB$

1965/6

In a plane a set of n points ($n > 3$) is given. Each pair of points is connected by a segment. Let d be the length of the longest of these segments. We define a diameter of the set to be any connecting segment of length d . Prove that the number of diameters of the given set is at most n .

Eight International Olympiad, 1965

1966/1

In a mathematical contest, three problems, A , B , C were posed. Among the participants there were 25 students who solved at least one problem each. Of all the contestants who did not solve problem A , the number who solved B was twice the number who solved C . The number of students who solved only problem A was one more than the number of students who solved A and at least one other problem. Of all students who solved just one problem, half did not solve problem A . How many students solved only problem B ?

1966/2

Let a , b , c be the lengths of the sides of a triangle, and α , β , γ , respectively, the angles opposite these sides. Prove that if

$$a + b = \tan \frac{\gamma}{2} (a \tan \alpha + b \tan \beta)$$

the triangle is isosceles.

1966/3

Prove: The sum of the distances of the vertices of a regular tetrahedron from the center of its circumscribed sphere is less than the sum of the distances of these vertices from any other point in space.

1966/4

Prove that for every natural number n , and for every real number $x \neq k\pi/2^t$ ($t = 0, 1, \dots, n$; k any integer)

$$\frac{1}{\sin 2x} + \frac{1}{\sin 4x} + \dots + \frac{1}{\sin 2^n x} = \cot x - \cot 2^n x.$$

1966/5

Solve the system of equations

$$|a_2 - a_1|x_2 + |a_1 - a_3|x_3 + |a_1 - a_4|x_4 = 1$$

$$|a_2 - a_1|x_1 + |a_2 - a_3|x_3 + |a_2 - a_4|x_4 = 1$$

$$|a_3 - a_1|x_1 + |a_3 - a_2|x_2 = 1$$

$$|a_4 - a_1|x_1 + |a_4 - a_2|x_2 + |a_4 - a_3|x_3 = 1$$

where a_1, a_2, a_3, a_4 are four different real numbers.

1966/6

In the interior of sides BC, CA, AB of triangle ABC , any points K, L, M , respectively, are selected. Prove that the area of at least one of the triangles AML, BKM, CLK is less than or equal to one quarter of the area of triangle ABC .

Ninth International Olympiad, 1965

1967/1

Let $ABCD$ be a parallelogram with side lengths $AB = a$, $AD = 1$, and with $\angle BAD = \alpha$. If $\triangle ABD$ is acute, prove that the four circles of radius 1 with centers A, B, C, D cover the parallelogram if and only if

$$a \leq \cos \alpha + \sqrt{3} \sin \alpha.$$

1967/2

Prove that if one and only one edge of a tetrahedron is greater than 1, then its volume is $\leq 1/8$.

1967/3

Let k, m, n be natural numbers such that $m + k + 1$ is a prime greater than $n + 1$. Let $c_s = s(s + 1)$. Prove that the product

$$(c_{m+1} - c_k)(c_{m+2} - c_k) \dots (c_{m+n} - c_k)$$

is divisible by the product $c_1 c_2 \dots c_n$.

1967/4

Let $A_0 B_0 C_0$ and $A_1 B_1 C_1$ be any two acute-angled triangles. Consider all triangles ABC that are similar to $\triangle A_1 B_1 C_1$ (so that vertices A_1, B_1, C_1 correspond to vertices A, B, C , respectively) and circumscribed about triangle $A_0 B_0 C_0$ (where A_0 lies on BC , B_0 on CA , and C_0 on AB). Of all such possible triangles, determine the one with maximum area, and construct it.

1967/5

Consider the sequence c_n , where

$$c_1 = a_1 + a_2 + \dots + a_8$$

$$c_2 = a_1^2 + a_2^2 + \dots + a_8^2.$$

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$$c_n = a_1^n + a_2^n + \dots + a_8^n.$$

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in which $a_1, a_2, \dots, a_8$ are real numbers not all equal to zero. Suppose that an infinite number of terms of the sequence c_n are equal to zero. Find all natural numbers n for which $c_n = 0$.

1967/6

In a sports contest, there were m medals awarded on n successive days ($n > 1$). On the first day, one medal and $1/7$ of the remaining $m - 1$ medals were awarded. On the second day, two medals and $1/7$ of the now remaining medals were awarded; and so on. On the n -th and last day, the remaining n medals were awarded. How many days did the contest last, and how many medals were awarded altogether?